

Epistemic Blind Spots in World-Model-Based Active Inference: Why RSSM Imagination Cannot See Obstacles

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Abstract

We investigate whether obstacle avoidance can emerge purely from Expected Free Energy (EFE) minimization in a deep active inference agent for autonomous driving—without privileged obstacle coordinates. Using an RSSM world model trained on CARLA highway scenarios, we systematically test whether learned preferences can drive obstacle avoidance through imagined trajectory planning. We discover a fundamental **epistemic blind spot**: while the RSSM posterior (filtering) correctly encodes obstacle presence—a contrastive preference model achieves 37 nats separation between obstacle-free and obstacle-present latent states—the RSSM prior (imagination) remains obstacle-blind. During CEM planning, all imagined trajectories produce similar latent sequences regardless of steering direction, because the world model cannot predict how obstacles enter or exit the visual field as a function of actions. This posterior-prior gap persists across three input modalities (64×64 RGB, 128×128 RGB, RGBD), multiple preference formulations (single GMM, contrastive log-ratio), and two training data strategies (collision-inclusive and collision-free obstacle demonstrations). Our results demonstrate that epistemic competence at the observation level (posterior) does not guarantee epistemic competence at the planning level (prior), identifying a critical limitation of current RSSM-based active inference for safety-relevant decision-making.

1 Introduction

Active inference [Friston, 2010, Parr et al., 2022] provides a unified framework for perception, learning, and action through free energy minimization. A key promise is that goal-directed behavior emerges from the agent’s generative model without hand-crafted reward functions: the agent imagines future trajectories, scores them by Expected Free Energy (EFE), and selects actions that minimize surprise relative to its preferences.

For this promise to hold in safety-critical domains like autonomous driving, the agent’s world model must be *epistemically competent*—it must know what it knows and, crucially, represent what it doesn’t know. When an obstacle appears on the road, the world model should encode this as a deviation from preferred states, and imagined trajectories should reflect whether actions resolve or worsen the situation.

We test this assumption using a deep active inference agent with a Recurrent State-Space Model (RSSM) [Hafner et al., 2023] world model, GMM preference distributions, and iCEM planning, evaluated on obstacle avoidance in CARLA 0.9.16 [Dosovitskiy et al., 2017]. Our contributions:

1. We identify an **epistemic blind spot** where the RSSM posterior correctly encodes obstacle presence but the prior (imagination) cannot predict obstacle dynamics from actions.
2. We introduce a **contrastive preference model** (log-ratio of two GMMs) that achieves 37+ nats separation between obstacle-free and obstacle-present posterior latents, demonstrating that the representation *contains* obstacle information.
3. We show this posterior-prior gap is **robust across modalities**: 64×64 RGB, 128×128 RGB, and RGBD (depth camera) all exhibit the same failure pattern.

4. We demonstrate that **data quality matters but is insufficient**: collision-free obstacle demonstrations improve world model driving quality (MLD 0.15m) but do not close the imagination gap.

2 Background

2.1 Active Inference and EFE

Under active inference, an agent selects policies π that minimize Expected Free Energy:

$$G(\pi) = \sum_t \gamma^t \left[\underbrace{\mathbb{E}_q[-\log p_{\text{pref}}(z_t)]}_{\text{instrumental}} - \underbrace{H[p(o_t|s_t)]}_{\text{epistemic}} \right] \quad (1)$$

where the instrumental term drives the agent toward preferred states and the epistemic term promotes information-seeking. For obstacle avoidance, if p_{pref} assigns low probability to obstacle-present states, the instrumental term should naturally drive evasive actions.

2.2 RSSM World Model

The RSSM maintains deterministic state $h_t \in \mathbb{R}^{256}$ (GRU) and stochastic state $z_t \in \mathbb{R}^{64}$ (Gaussian):

$$\text{Posterior: } z_t \sim q(z_t|h_t, o_t) \quad (\text{filtering}) \quad (2)$$

$$\text{Prior: } \hat{z}_t \sim p(z_t|h_t) \quad (\text{imagination}) \quad (3)$$

During **filtering** (observation available), the posterior incorporates visual evidence—including obstacles. During **imagination** (planning), only the prior is available, conditioned on the deterministic state and actions. The epistemic blind spot arises when the posterior encodes information that the prior cannot reproduce.

3 Experimental Setup

Architecture. ConvEncoder (4-layer CNN, $64 \times 64 \rightarrow 256\text{D}$), RSSM (deter=256, stoch=64), ObsDecoder, StateDecoder, 5-head ensemble for epistemic uncertainty. iCEM planner (500 samples, 50 elites, 5 iterations, horizon 15).

Tasks. CARLA 0.9.16 highway: Task A (lane keeping, Town04, moderate curves) and Task B (obstacle avoidance, Town06, 2–3 static obstacles per route, 373–518m).

Training data. 96K frames obstacle-free expert driving (Town04 + Town06) plus 10–30K frames obstacle-encounter data (BehaviorAgent and scripted lane-change demonstrations).

Privileged baseline. Task B with runtime ground-truth obstacle coordinates achieves 100% avoidance (28/28 obstacles) via reflexive steer prior, motor primitives, and directional CEM penalty. All six privileged mechanisms are removed for the pure AIF experiments.

4 Results

4.1 The Preference Gap is Inverted

We fit a GMM preference model on obstacle-free expert latents and evaluate log-probability on held-out obstacle-free vs. obstacle-present latents. If the GMM correctly encodes “prefer obstacle-free states,” obstacle-present latents should have lower log-probability.

The gap is **inverted in all cases**: obstacle-present latents receive *higher* GMM probability. The RSSM latent space encodes driving dynamics (speed, steering, lane position) rather than scene content. Higher resolution (128×128) narrows the gap but does not reverse it. Adding depth worsens it.

Table 1: Preference gap: $\Delta = \log p(z_{\text{clean}}) - \log p(z_{\text{obs}})$. Positive = correct direction.

Input modality	Δ (nats)	Recon ratio
64×64 RGB	−24.6	1.62×
128×128 RGB	−2.3	1.91×
64×64 RGBD	−68.1	1.35×

Table 2: Contrastive preference: posterior separation is correct, but obstacle avoidance remains at 0%.

Scale α	Raw gap	MLD (m)	Avoidance
1.0	+37.0 nats	4.0	0% (off-road)
0.3	+9.2 nats	0.32	0%
0.2	+1.1 nats	0.17	0%
0.1	+1.0 nats	0.31	0%

4.2 Contrastive Preference Achieves Posterior Separation

We replace the single GMM with a contrastive model: $\log p_{\text{pref}}(z) = \log p_{\text{clean}}(z) - \alpha \cdot \log p_{\text{avoid}}(z)$, where p_{clean} is fitted on obstacle-free and p_{avoid} on obstacle-present latents.

At $\alpha=1.0$, the 37 nats gap is strong but overwhelms driving (MLD 4.0m). At $\alpha \leq 0.3$, driving is preserved but avoidance remains 0%. The contrastive signal correctly separates *posterior* latents but cannot influence *imagined* trajectories.

4.3 The Epistemic Blind Spot

The root cause: during CEM planning, the RSSM imagines 500 trajectories via `img_step` (prior only). These imagined latent sequences do not encode obstacle proximity because the prior cannot predict obstacle appearance/disappearance from steering actions. All trajectories—straight, left, right—produce similar latent sequences in imagination.

Evidence: The 128×128 RGB model achieves 100% success rate (goal reached) with 0.17m mean lateral deviation and zero offroad events—the best driving quality of any variant—yet avoids zero obstacles. The agent drives perfectly but is blind to obstacles in its plans.

4.4 Clean Data Does Not Close the Gap

We collected 30K collision-free obstacle avoidance demonstrations (40% clean rate via waypoint-redirect autopilot) and retrained. With 10K obstacle frames (9.4% of training data), driving quality is preserved (MLD 0.15m) but avoidance remains 0%. The RSSM learns to *reconstruct* obstacle images (1.62× higher reconstruction error) but not to *imagine* obstacle dynamics.

5 Discussion

5.1 Posterior vs. Prior Epistemic Competence

Our results reveal a dissociation between two levels of epistemic competence in world-model-based active inference:

Posterior competence (filtering): The RSSM posterior correctly encodes obstacle presence when conditioned on observations. The contrastive preference achieves 37+ nats separation. Reconstruction error is 1.62× higher for obstacle images. The information *is there*.

Prior incompetence (imagination): The RSSM prior cannot predict how obstacle-related visual features evolve as a function of steering actions. During open-loop rollouts, the stochastic latent reverts to a dynamics-dominated representation where obstacle presence is lost.

This posterior-prior gap is an **epistemic blind spot**: the agent can recognize danger when it sees it but cannot predict danger from its plans. In active inference terms, the generative model’s *likelihood* (observation model) encodes obstacles, but the *transition* model (dynamics) does not propagate this information through action-conditioned rollouts.

5.2 Implications for AIF Safety

This finding has direct implications for deploying active inference agents in safety-critical settings:

1. **EFE planning is only as good as imagination.** A preference model that perfectly separates safe from unsafe states is useless if the world model cannot imagine the consequences of actions in those states.
2. **Observation-level epistemic uncertainty is insufficient.** Ensemble disagreement and reconstruction error detect obstacles at the observation level, but these signals are not available during imagination (no observations).
3. **The latent bottleneck prioritizes dynamics over content.** With 64 stochastic dimensions, the RSSM encodes temporally smooth driving dynamics rather than spatially localized scene content (obstacles). This is efficient for lane keeping but blind to safety-critical objects.

5.3 Toward Closing the Gap

Three directions may address this epistemic blind spot while remaining within the active inference framework:

Larger latent space. Increasing stochastic dimensions from 64 to 256 may provide enough capacity to encode both driving dynamics and scene content. This is the most self-supervised approach but may require proportionally more training data.

Auxiliary prediction signal. A supervised obstacle detection head during training forces the latent space to encode obstacle presence. At evaluation time, only the shaped latent representation is used—no privileged information. This is analogous to learning with privileged information (LUPI).

Object-centric architectures. Models like SLATE [Singh et al., 2022] or SAVi [Kipf et al., 2022] that decompose scenes into object slots may naturally separate obstacles from background, making obstacle dynamics explicitly representable in imagination.

6 Related Work

Deep active inference. Catal et al. [2020] and Fountas et al. [2020] apply active inference with VAE world models to visual navigation and Atari. Millidge [2020] connects active inference with policy gradients. Lanillos et al. [2021] surveys active inference in robotics. Our work extends to continuous-action driving and identifies a specific failure mode in RSSM-based imagination.

World models for driving. DreamerV3 [Hafner et al., 2023] uses RSSM imagination for RL-based policy learning. MILE [Hu et al., 2022] learns driving world models for imitation. Our finding—that RSSM imagination is obstacle-blind—may apply to these systems as well.

Epistemic uncertainty in planning. Chua et al. [2018] use ensemble disagreement for model-based RL. Sekar et al. [2020] explore planning to explore. Our work shows that epistemic signals available during filtering (posterior) may not transfer to planning (prior).

7 Conclusion

We identified a fundamental epistemic blind spot in RSSM-based active inference: the world model’s posterior correctly encodes obstacle presence (37 nats contrastive separation), but the prior (imagination) cannot predict obstacle dynamics from actions. This posterior-prior gap persists across input modalities (RGB, high-resolution RGB, RGBD), preference formulations (single GMM, contrastive), and training data quality

(collision-inclusive, collision-free demonstrations). The result is an agent that drives excellently (100% success rate, 0.17m lateral deviation at 128×128) but is completely blind to obstacles in its planning horizon.

This finding highlights that epistemic competence at the observation level does not guarantee epistemic competence at the planning level—a critical consideration for deploying active inference agents in safety-relevant domains. Closing this posterior-prior gap, through larger latent representations, auxiliary training signals, or object-centric architectures, is essential for realizing the full promise of active inference for autonomous decision-making.

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